

DATA TREATMENT PLATFORM

ELT Pipeline Architecture

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Context & Goals

- **Project Goal:** Build a scalable Data Platform for BigMedia.
- **Problem:** Need to process large volumes of viewing logs and user data.
- **Objective:** Enable data-driven decision making via PowerBI dashboards.

Business Context Image

Why this platform?

- Centralize data from multiple sources.
- Ensure data quality and consistency.
- Provide near real-time insights.

Data Sources Generated by Faker

Source	Description	Format
Contents	Catalog of movies and series	CSV
Users	Registered users data	CSV
Viewing Logs	User activity logs (High Volume)	JSON
Social Media	Mentions and interactions	JSON

Faker Data Builder: Simulates realistic production data with referential integrity and temporal consistency.

ARCHITECTURE & INFRASTRUCTURE



The Philosophy

- **Modern ELT:** Load raw data first, transform in-warehouse.
- **Orchestrator:** **Prefect** manages dependencies.
- **Platform:** 100% Serverless on GCP.

Key Components

- **Faker:** Simulates production inputs.
- **dbt:** Medallion architecture transformations.
- **Terraform:** Fully reproducible Infra.

1. Developer Experience (Mise)

- **Tooling:** Manages versions for Terraform, Python, and dbt via `mise.toml`.
- **Templating:** Ensures consistent environment setup across the team.
- **No “Works on my machine”:** Unified config.

3. Infrastructure (Terraform)

Provisioning on GCP:

- **State:** Remote state in GCS (Collaboration).
- **Security:** IAM Roles defined in code.
- **Environments:**
 - dev: Sandbox for testing.
 - prod: Live data pipeline.

Infrastructure is treated exactly like software.

2. Version Control (Git & GitHub)

- **Tracking:** Full history of code changes.
- **CI/CD:** GitHub Actions triggers on Pull Requests.
 - Linting (SQLFluff, Black).
 - dbt compile checks.

INGESTION & TRANSFORMATION

Prefect Flow: `refill_bucket_flow`

1. Generator (`seed_script.py`)

- `contents.csv` (Catalog)
- `users.csv` (User Base)
- `viewing_logs.json` (High Vol.)
- `mentions.json` (Unstructured)

2. Data Lake (GCS)

- Bucket: `raw_data/ingested-data`
- **Partitioning:** By Date/Run
- **Safety:** Idempotent execution

Key Technical Choice: Simulating a live production environment rather than static file uploads.

The “Medallion” Architecture in BigQuery



Layer Details

- **Staging:** 1:1 Views, Renaming.
- **Intermediate:** Complex joins (e.g., `int__viewing_enriched`).
- **Marts:** Final Business Logic.

Why External Tables? We query GCS directly using `create_external_sources`.

- No data duplication.
- Lower storage costs.

INGESTION & TRANSFORMATION

Workflow: Faker -> GCS

- **Generation:** `seed_script.py` creates synthetic data.
- **Storage:** Uploaded to GCS `raw_data/` bucket.
- **Partitioning:** Organized by execution timestamp.
- **Idempotency:** Prevents data duplication on re-runs.

Ingestion Flow Diagram

Medallion Architecture

Bronze (Raw) -> Silver (Staging/Int) -> Gold (Marts)

Bronze

Raw ingestion via
External Tables

Silver

Cleaning,
deduplication,
enrichment [\[1\]](#)

Gold

Star Schema (Facts
& Dimensions) for
BI

ANALYTICS & VISUALIZATION

Star Schema for Reporting

- **Fact Table:**
 - fct__viewings
- **Dimension Tables:**
 - dim__users
 - dim__content
 - dim__devices
 - dim__date

**Entity Relationship Diagram
(ERD)**

Visualizing the Insights

PowerBI Dashboard Screenshot

- **Key KPIs:** Total views, active users, content performance.
- **Interactivity:** Filters by date, device, and content type.
- **DirectQuery:** Connected directly to BigQuery Gold layer [\[2\]](#).

Thank You!

Questions?

Built with: Prefect, GCP, dbt, PowerBI

REFERENCES

- [1] dbt Labs, “Data Build Tool.” [Online]. Available: <https://www.getdbt.com/>
- [2] “Microsoft Power BI.” [Online]. Available: <https://powerbi.microsoft.com/>